

The UK Linguistics Olympiad 2017



Solution and marking.

Scoring (max 31)

- 5.1: 2 points for every correctly-filled cell. (max 16)
- 5.2: 1 point for every correct Yup'ik word. (max 5)
- 5.3: (max 10)
 - 3 points for a mathematical explanation of the magic square.
 - 5 points for a linguistic explanation of the Yup'ik number system.
 - 2 points for an explanation of how the clue numbers lead to the solution.

5.1

	1	2	3
a	4	9	2
b	3	5	7
c	8	1	6

5.2 Yuinaat yuinaq malruk akimiaq atauciq.

5.3 Explanation: see the commentary.

The UK Linguistics Olympiad 2017



Commentary (by the problem's author, Kai Low Rui Hao)

Candidates should recognize this as a mathematics problem and take into consideration the possibility of the spelling of numbers being represented in a different base (for example, in French, 99 is *quatre-vingt dix neuf*, which translates to $4 \times 20 + 10 + 9$).

There is only one possible type of 3-by-3 magic square, although there are 8 distinct configurations that stems from the one type due to rotation and reflection. These permutations are avoided due to the presence of the Yup'ik cross-number puzzle clues.

Since there are 9 digits, the numbers can be arranged via this formula

N-1	N+4	N-3
N-2	N	N+2
N+3	N-4	N+1

This formula takes into consideration 9 consecutive numbers, as well as the fact that the sum of every row, columns, and diagonals is the same – $3n$. Since the sum of each column is $3n$, the sum of three columns is $9n$. The sum from 1 to 9 is 45. Hence $9n = 45$, leaving us with $n=5$.

The answer to Task 1 is as above. Candidates may in the process arrive at the other 7 possible combinations. These permutations will be eliminated when checked against the yup'ik hints.

Since the question provided 2Down as a 3-digit number that starts with 9, and all numbers in the clues are 3-digit, and all Yup'ik numbers begin with Yuinaat, candidates have to consider that Yup'ik may be using a base larger than 10. The most common, in fact, is the Vigesimal system, which is base 20. This is also practical based on the background provided in the question – Yup'ik people based their concept of counting on body parts (20 fingers + toes).

Despite the various permutations, candidates should be able to arrive at 2D, which is 951. Further attempts at solving the spelling will reveal that $951 = (20 \times 20 \times 2) + (20 \times 7) + (10 + 1)$. This is true to the base 20 system. Arriving at this conclusion will reveal:

1. Suffix -k: multiply by 2/double/to do with two (a dual number)
2. Suffix -t: multiply by more than 2 (a plural number)
3. Suffix -q: the root suffix.

If the word ends in suffix -q, it signifies to the candidate that the subsequent number should be an addition and not a multiplication.

Candidates can then work on 3Across, which is the next biggest number with $20 \times 20 \times 2$. Hence, the number, which is the second biggest after one starting with "9", should start with "8". This will lead to 3A being 816. Candidates can then associate *akimiaq atauciq* to 16. From the 2D, *atauciq* = 1, hence *akimiaq* is 15. This is a reasonable and valid guess since Yup'ik people pay attentions to numbers based on the hands and feet.

The UK Linguistics Olympiad 2017



Next, refer to 3D where candidates can make out $(20 \times (10+?)) + 16$. Given that diagonals also sum up to 15, we can gather 2 and then 7 from it being the remaining cell in 3D. Hence 3D is 276 and it is $(20 \times (10+3)) + 16$.

Knowing that *qula* = 10, *akimiaq* = 15, and that *malruk* = 2, *pingayun* = 3, we will be able to solve most of the magic square. The other number not mentioned is *cetaman* = 4.

Task 2: Yuinaat yuinaq malruk akimiaq atauciq.

The diagonal 1a-2b-3c is 456 and can be expressed in Yup'ik spelling as $(20 \times 20) + (20 \times 2) + 16$. However, notice from other clues that numbers below 800 are not phrased in this manner but more of $(20 \times 22) + 16$. Hence the number will be expressed as $(20 \times (20 + 2)) + 16$.